How do mRNA vaccines work?

Summary — An mRNA vaccine gives cells a short-lived recipe for one pathogen protein, not the pathogen itself. A lipid shell gets the recipe into cells and also helps alert the immune system; antigen production then recruits antibodies, memory B cells and T cells. The platform has delivered strong protection from severe respiratory disease, but protection from infection wanes, variant matching is imperfect, and myocarditis is a rare risk concentrated in young males after dose 2.

The platform is a transient, programmable protein factory

- The payload is messenger RNA (mRNA) inside a lipid nanoparticle (LNP): the particle protects RNA, promotes cell entry and cytoplasmic release, and ribosomes translate the sequence without the RNA entering the nucleus or becoming DNA. Chaudhary et al. 2021, Hou et al. 2021, Pardi & Krammer 2024
- Ionizable lipids bind RNA during particle formation but are closer to neutral in the body, improving delivery over permanently charged carriers. Cullis & Felgner 2024, Hou et al. 2021
- The encoded antigen is designed: COVID-19 vaccines used prefusion-stabilized spike, building on earlier coronavirus structural work, so new targets can reuse the manufacturing platform. Corbett et al. 2020, Pardi & Krammer 2024

Modified RNA and purification made useful expression possible

- Unmodified RNA activates innate sensors: substituting modified nucleosides suppressed signalling through Toll-like receptors 3, 7 and 8 and reduced cytokine release from human dendritic cells. Karikó et al. 2005
- Removing double-stranded RNA contaminants by high-performance liquid chromatography eliminated detectable interferon and inflammatory-cytokine induction in the tested cells and raised translation **10- to 1,000-fold**. Karikó et al. 2011
- Translation fidelity is a real design constraint: N1-methylpseudouridine caused +1 ribosomal frameshifting and immune recognition of off-target products in mice and people, although no adverse outcome was reported and synonymous recoding reduced the signal. Mulroney et al. 2024

The carrier is both delivery vehicle and immune alarm

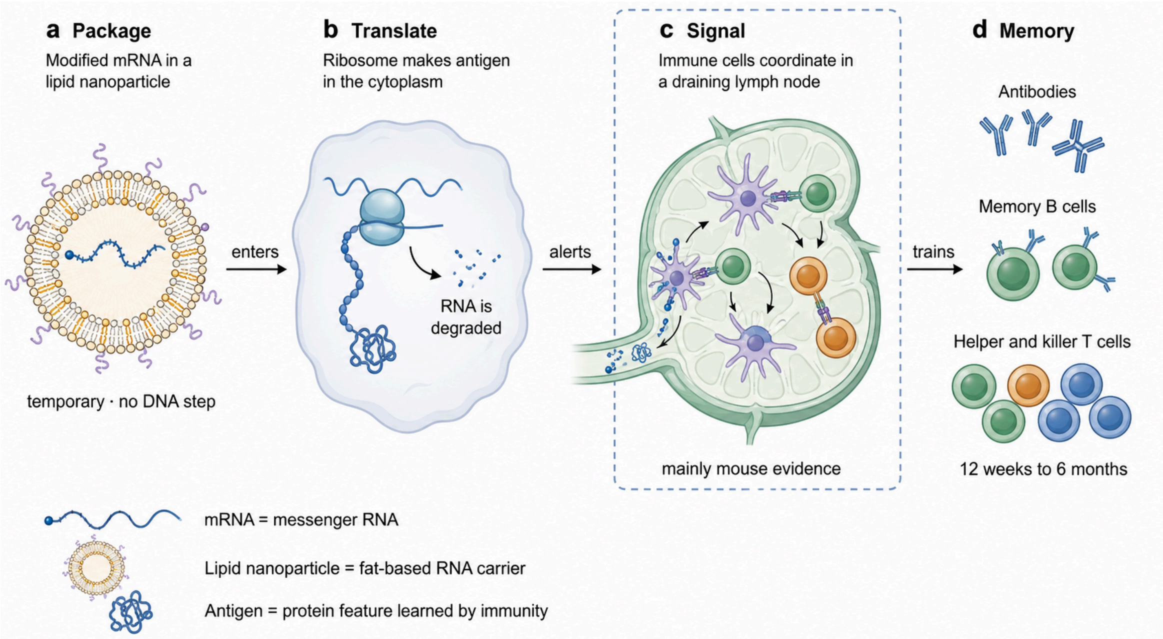
- In mice, the tested LNP formulation acted as an adjuvant even without antigen-encoding RNA, driving T-follicular-helper and germinal-centre responses through its ionizable lipid and interleukin-6; the exact pathway and magnitude cannot be assumed for every LNP or for humans. Alameh et al. 2021, Verbeke et al. 2022
- Mouse tissue profiling found vaccine mRNA chiefly in draining-node monocytes, macrophages and migratory dendritic cells at day 1; MDA5-type I interferon signalling supported antigen-specific CD8 T-cell expansion, while several other proposed pathways were dispensable in that model. Li et al. 2022

Lymph nodes convert a brief signal into durable immune memory

- Early human trials found strong neutralizing antibodies and TH1-skewed CD4 and CD8 T-cell responses, while CD8 cells expanded after the first dose before neutralizing antibodies had fully matured. Sahin et al. 2020, Oberhardt et al. 2021

- Serial lymph-node sampling found spike-specific germinal-centre B cells for at least **12 weeks** and T-follicular-helper cells for about **6 months**, although these intensive studies involved small, relatively young cohorts. Turner et al. 2021, Mudd et al. 2022

- Across 1,540 tracked B-cell clones, somatic mutation rose **3.5-fold over six months**; affinity-matured memory B cells and bone-marrow plasma cells persisted even as circulating antibody declined. Kim et al. 2022



the evidence-supported route from injection to memory. Delivery, transient expression and persistent human lymph-node responses are observed; detailed sensing pathways and LNP adjuvanticity rest mainly on mouse experiments. Pardi & Krammer 2024, Alameh et al. 2021, Li et al. 2022, Turner et al. 2021, Kim et al. 2022

Antibodies predict protection, but platform performance depends on pathogen and outcome

- The pivotal placebo-controlled COVID-19 trials reported **95.0%** efficacy (95% credible interval 90.3-97.6; n=43,448) for BNT162b2 and **94.1%** (95% CI 89.3-96.8; n=30,420) for mRNA-1273 against symptomatic disease before major immune-escape variants; six-month BNT162b2 efficacy was 91.3% (89.0-93.2) and 96.7% (80.3-99.9) against severe disease. Polack et al. 2020, Baden et al. 2021, Thomas et al. 2021

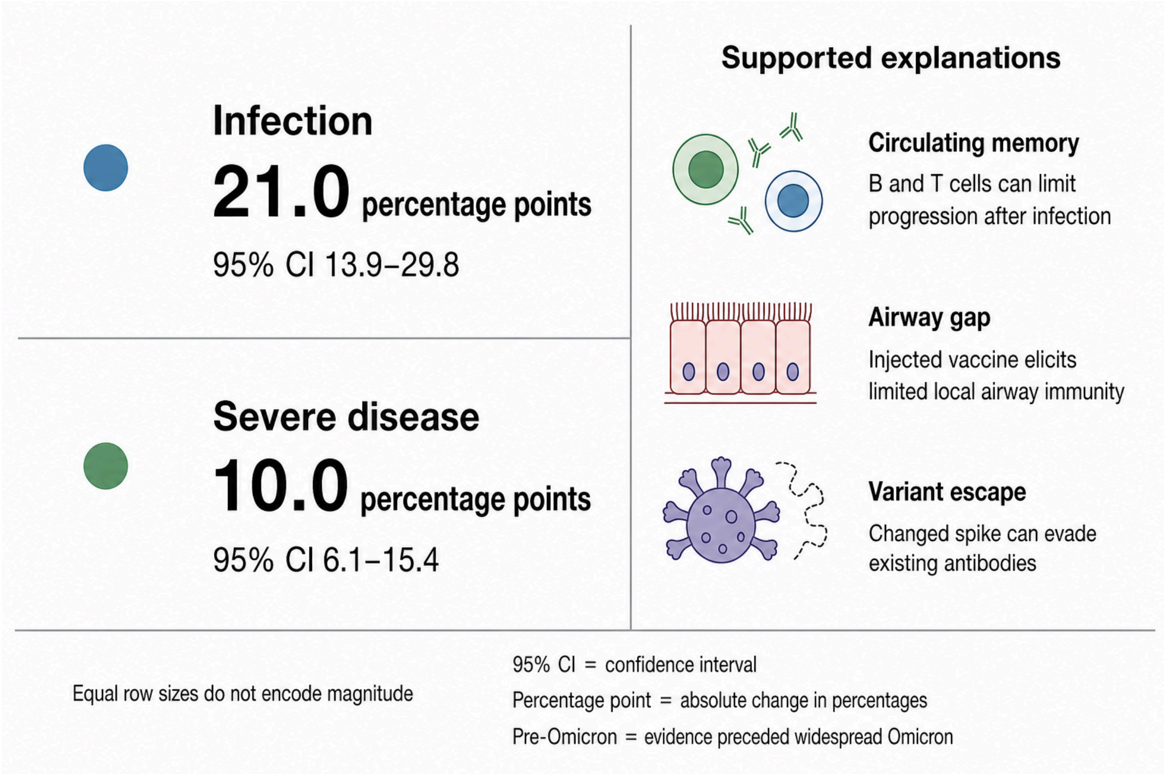
- In mRNA-1273 recipients, estimated vaccine efficacy rose from 78% (95% CI 54-89) at a neutralizing titre of 10 to 96% (94-98) at 1,000; a 24-study meta-analysis also linked neutralization loss to reduced protection across variants. Correlation does not mean antibodies mediate every component of protection. Gilbert et al. 2022, Cromer et al. 2022

- The platform generalizes but does not guarantee success: one-dose mRNA-1345 prevented RSV lower-respiratory disease by 83.7% (95.88% CI 66.0-92.2; n=35,541), whereas an influenza formulation produced stronger A but weaker B responses than licensed vaccine. Wilson et al. 2023, Ma et al. 2025, Kandinov et al. 2025, Whitaker et al. 2023

VACCINE	EVIDENCE	MAIN RESULT	IMPORTANT BOUNDARY
BNT162b2	Placebo RCT, n=43,448	95.0% efficacy against symptomatic COVID-19	Ancestral-virus era Polack et al. 2020
mRNA-1273	Placebo RCT, n=30,420	94.1% efficacy; 30 severe cases, all placebo	Ancestral-virus era Baden et al. 2021
mRNA-1345 RSV	Placebo RCT, n=35,541, age at least 60	83.7% against lower-respiratory disease with at least 2 signs	Median follow-up 112 days Wilson et al. 2023
ARCT-154 self-amplifying RNA	Placebo RCT, 5 micrograms	56.6% against any and 95.3% against severe COVID-19	Mostly Delta; short follow-up Hô et al. 2024

Protection wanes fastest where the airway and variants matter most

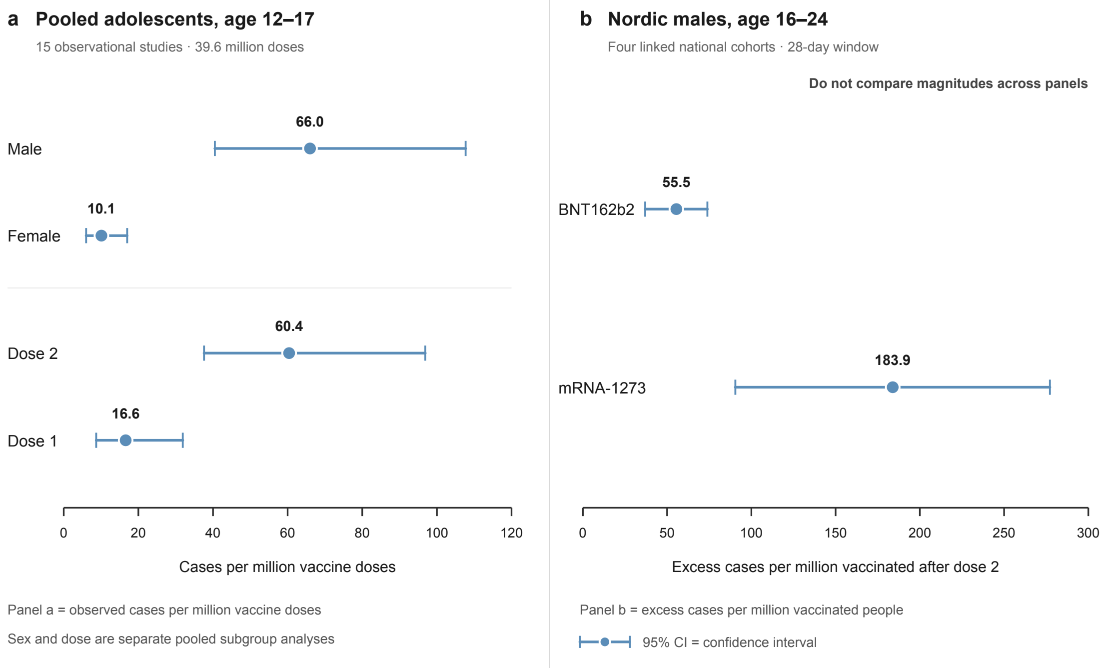
- Across 78 estimates, effectiveness fell **21.0 percentage points** (95% CI 13.9-29.8) against infection from month 1 to 6, versus **10.0 points** (6.1-15.4) against severe disease; only 3 of 18 included studies had low overall risk of bias. Feikin et al. 2022
- The anatomical pattern fits: vaccinated people had strong blood responses but lower airway neutralization and no detectable spike-specific B or T cells in bronchoalveolar lavage, unlike convalescents. This small study supports, but does not by itself prove, a mechanism for breakthrough infection. Tang et al. 2022
- Variant matching helps incompletely. A BA.1-bivalent booster raised BA.1 titres over the ancestral booster but did not test effectiveness; later BQ.1.1 and XBB.1 escaped much BA.5-bivalent-dose neutralization. Chalkias et al. 2022, Kurhade et al. 2023
- A 28-study synthesis found bivalent boosters added about **28-31% relative effectiveness** against infection and **58-62%** against severe outcomes versus earlier-dose comparators, with substantial heterogeneity. Song et al. 2024
- XBB.1.5-vaccine effectiveness in 53.4 million adults fell by month 5 to 26.7% against infection, 52.3% against hospitalization and 69.4% against death; JN.1 replacement reduced it further. Antibody responses remained biased toward epitopes shared with the ancestral strain, with limited truly XBB-specific recruitment in some people. Ma et al. 2026, Johnston et al. 2024



waning is outcome-dependent. The two estimates and 95% confidence intervals come from a pre-Omicron systematic review/meta-regression; equal row sizes deliberately avoid encoding a false visual scale. Airway sampling and later variant studies explain why infection protection is more fragile, but they do not partition the decline quantitatively. Feikin et al. 2022, Tang et al. 2022, Kurhade et al. 2023, Ma et al. 2026

Myocarditis is rare, real, and sharply concentrated by age, sex, dose and product

- Active surveillance of 11.8 million mRNA doses found no prespecified serious-outcome signal and anaphylaxis at about **5 per million doses**; a separate matched national study found most examined adverse events were not elevated but confirmed a myocarditis excess. Klein et al. 2021, Barda et al. 2021
- A 99-million-person multinational study reproduced the myocarditis/pericarditis signal after mRNA vaccination; its observed-versus-expected design remains sensitive to background-rate assumptions. Faksova et al. 2024
- In 39.6 million adolescent doses, pooled myopericarditis was **43.5 per million** (95% CI 30.8-61.6): **66.0** in males versus **10.1** in females, and **60.4** after dose 2 versus **16.6** after dose 1. Guo et al. 2023
- Nordic registers showed that males aged 16-24 had **55.5 excess cases per million** after BNT162b2 dose 2 and **183.9 per million** after mRNA-1273 dose 2; English data likewise found the highest relative risk after mRNA-1273 dose 2 and a smaller booster signal. Karlstad et al. 2022, Stowe et al. 2023
- Infection is not a uniform comparator: across all ages, infection-associated myocarditis risk was higher, but among men younger than 40 the estimated excess after mRNA-1273 dose 2 was 97 per million versus 16 per million after a positive test. Patone et al. 2022



the signal is heterogeneous, not a single platform-wide rate. Panel A pools observational adolescent studies; panel B uses Nordic register-derived excess events in young men. Different populations, denominators and axes make within-panel comparisons valid but cross-panel marker positions non-comparable. Guo et al. 2023, Karlstad et al. 2022, Patone et al. 2022, Faksova et al. 2024

Speed and programmability remain constrained by stability, delivery and evidence gaps

- Purification, particle mixing, scale-up and cold-chain stability remain bottlenecks. Rosa et al. 2021, Hou et al. 2021
- Freeze-drying preserved one mRNA-LNP formulation for 12 weeks at room temperature and 24 weeks at 4 degrees C in preclinical tests, but another study found post-drying RNA retention and translation depended strongly on the ionizable lipid; thermostability is therefore formulation-specific. Muramatsu et al. 2022, Lamoot et al. 2023
- Self-amplifying RNA includes replication machinery and can reduce dose: 5-microgram ARCT-154 protected against severe Delta-era COVID-19, and as a fourth-dose booster it produced non-inferior ancestral and higher BA.4/5 neutralization than BNT162b2 at day 28; neither result establishes long-term superiority. Blakney et al. 2021, Hô et al. 2024, Oda et al. 2024

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